RESTRICTED SERVICE NOTE

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FILE TITLE Signa® Images/Artifacts SN 65032

PERIPHERAL SIGNAL ARTIFACTS

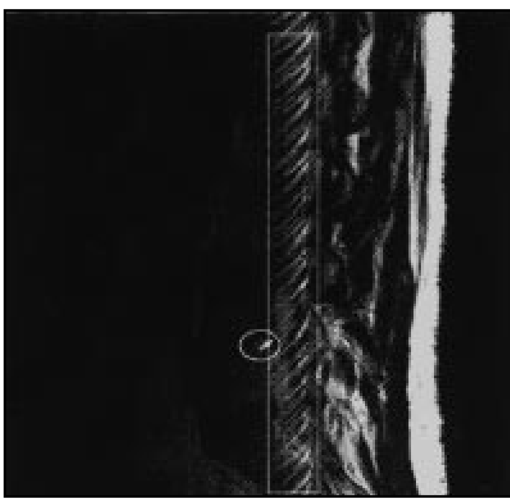
Purpose: To explain the source of some of the image artifacts that can occur on normally functioning systems. These artifacts are not due to defective hardware or software, but due to limitations in the MR physics and applications.

Peripheral signal artifacts can appear as either a bright spot or as a ribbon of signal smeared through the image. We sometimes refer to these kinds of artifacts as Startifacts (bright spot) or Annefacts (ribbon). Basically both of theses artifacts have the same root cause. Signals are generated outside the desired field of view (FOV) and the receiver is able to detect them.

Startifact signals, which appear as a bright star close to the middle of the image, have their origins very far from isocenter. In that non-linear region the FID (free induction decay) signal coming off the RF 180 pulse or from a SAT pulse isn't crushed out and it aliases back into the image. (Aliasing is caused primarily by the fact that our gradient and magnetic coils have finite lengths, resulting in the gradient and B_0 fields becoming increasingly non-linear away from iso-center.) A similar artifact also occurs in 3-D gradient PSDs for essentially the same reason.

Annefact appears in Fast Spin Echo (FSE) scans as a small bright ghost smeared through the image in the phase direction. The origin is also far from iso-center where the gradients are non-linear, and is combined with the B_0 (magnetic) field's inhomogenity to produce a spatially compressed (therefore very bright) spin echo signal which is alaised back into the image. Uncompensated eddy currents in this area cause phase errors in the compressed signal and smear it through the image.

Example: One of the most common complaints occurs during Sagittal (and Coronal) spine scans ilsing a 4-channel phase array coil with Fast Spin Echo (FSE). The artifact occurs when phase encoding is in the S/I (Z-axis) direction and No Phase Wrap is turned on. (The reason for swapping phase and frequency (SPF), to phase encode in the S/I direction, is to reduce CSF flow related artifacts. No Phase Wrap (NP) is selected to prevent the anatomy, which extends outside the Field of View (FOV), from alaising back into the image in the S/I direction.) The image below has 2 distinct artifacts, a Startifact (oval) and an Annefact (rectangle).



|- - - - - - - - - - - Image Parameters - - - - - - - - - - - - -|
Image .....3 / 9 Tr 3000.00 ms
Echo ....1/1 Te 102.00 ms
Plane Sagittal
SPF....Yes
PSI) name: fse

Slice Thickness. 4.00 mm NEX 2.00
Slice Spacing... 1.00 mm **Coil CTLBOT**
Scan Time.. 3: 18.10 ETL.. 8
BandWidth.. 16.0 Khz

Scan Matrix 256 x 256 **FOV 26.00 cm**
Scan Options: FC ST **NP**
SAT Pulses : SI

Workaround By selecting the receive coils that match the imaging FOV (i.e. LS45, LS56, CS12, etc.), you can lessen the likelihood that you will pickup the peripheral signals that are generated outside the field of view.

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Figures One and Two show simulated and ideal plots of frequency versus distance along the Z-axis for the magnetic and gradient fields. Figure Three combines both these fields, and is important in understanding what is happening when alaising occurs.

In Figure Three there is a peak in the simulated plot sometimes referred to as the turn around point". Note that some frequencies are repeated as one moves beyond the turn around point. This repetition in frequencies is the root cause of these artifacts - essentially there are multiple points in space with the same B_0.

Note: The turn around point will vary depending on magnet (i.e. S1, S2, S3, etc.) and gradient coil types. These plots shown here are for discussion purposes only and do not show the full complexity of the issue. Other factors that contribute to the issue are B1, T2* and eddy currents.

FIGURE ONE - B0 FREQUENCY VS. DISTANCE (7-axis only)
ITypical Scanning Range +/- 24cm I

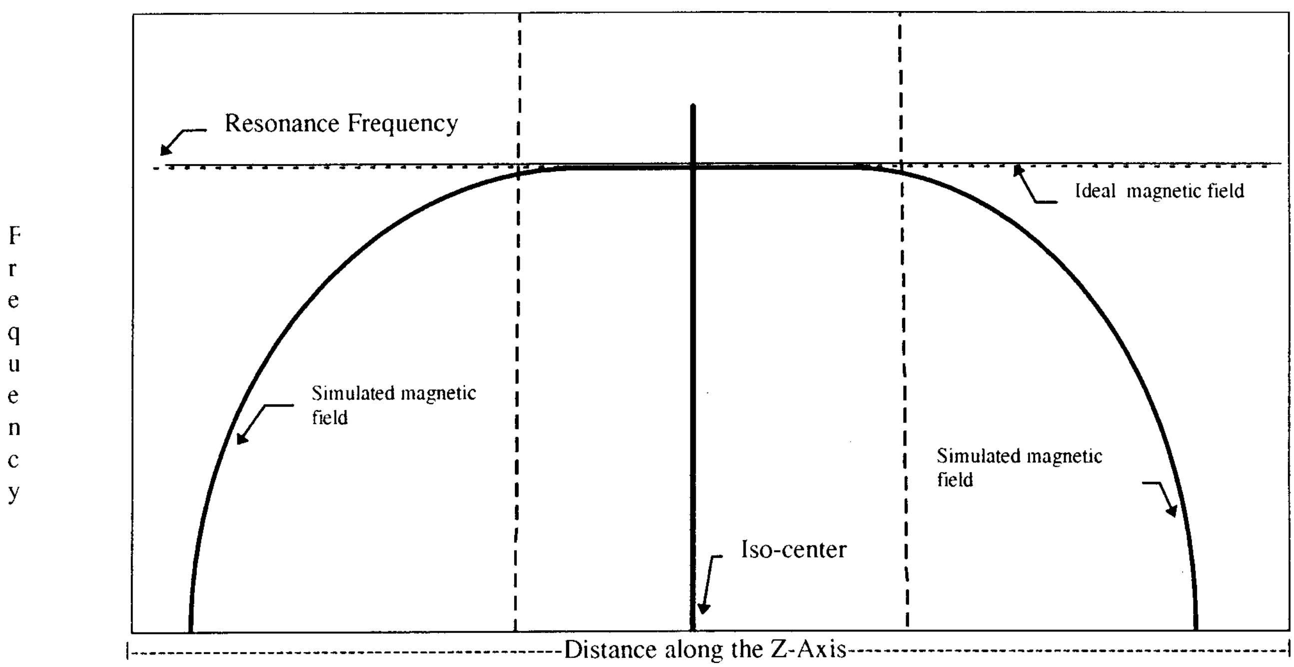

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FIGURE TWO - GRADIENT FIELD FREQUENCY VS. DISTANCE (Z-axis only)

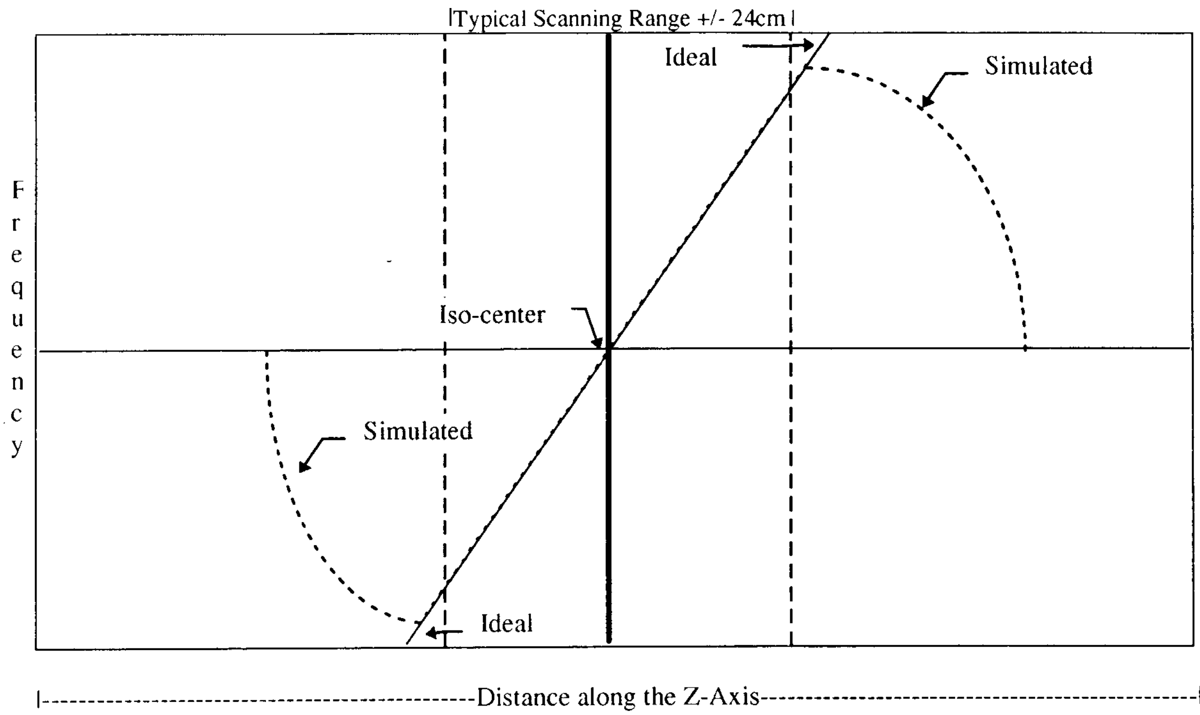
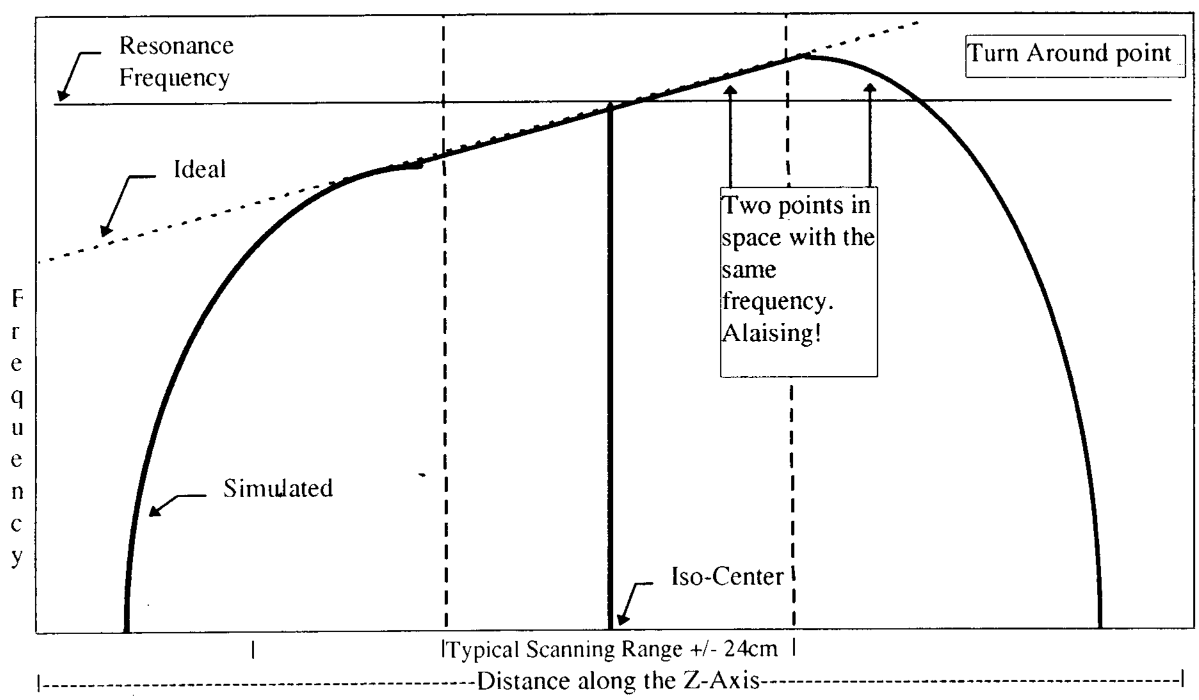


FIGURE THREE - B0 PLUS GRADIENT FIELD FREQUENCY VS. DISTANCE (Z-axis only)



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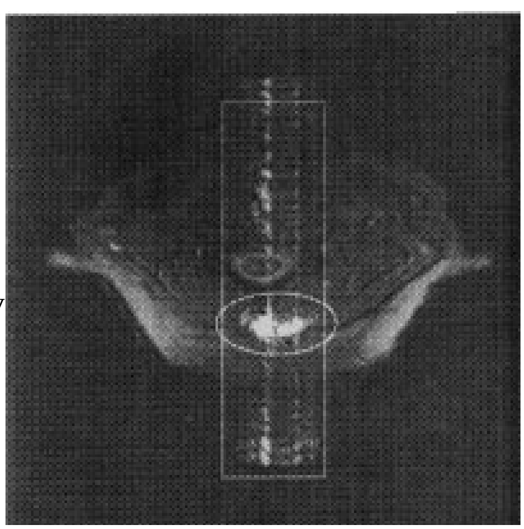
OTHER EXAMPLES:

Annefact. 5.4 system. FSE optimization lightened the artifact, but didn't completely remove it.
Workaround: Use CS12 and do not swap phase and frequency.

| - - - - - - - - - - - - Image Parameters - - - - - - - - - - - - |
Image........ 6 / 24 Tr 2962.96 ms
Echo .... 2/2 Te 102.00ms
Plane **Sagittal** Ti 0.00 ms
SPF....**Yes** Contig.. 1 Flip.... 90 deg.
Slice Thickness. 3.00 mm NEX 2.00
Slice Spacing... 1.00 mm Coil **CTLTOP**
ScanTime.. 6:25.29 ETL.. 8 RawNo..9218
BandWidth..32.0 Khz Ave Sar..0.70 Pk Sar 1.39

Study Number.... 1029 Series 2
Scan Matrix 512 x 256 FOV **24.00 cm**
Scan Options: ST PG NP
SAT Pulses :
PSD name: **fse**

Startifact and Annefact with 3dgre scan. Signal alaising from signals received from coils 3&4.
Workaround: Use CS1 2 instead of CTLTOP



|- - - - - - - - - - - - Image Parameters - - - - - - - - - - - -|
Image..... 28 / 60 Tr 42.00 ms
Echo ...... 1 / 0 Te 15.00 ms
Plane Axial Ti 0.00 ms
SPF.... .... No Contig.. 1 Flip ..... 5 deg.
Slice Thickness. 2.00 mm NEX 1.00
Slice Spacing... 0.00 mm Coil **CTLTOP**
ScanTime.. 4:21.62 ETL.. 0 RawNo..48157
BandWidth.. 16.0 Khz Ave Sar..0.47 Pk Sar..0.94
Scan Matrix 256 x 128 FOV **20.00 cm**
Scan Options: FC ST
SAT Pulses :
a: Loc:50.0:Width:80.0
p:Loc:6493.6:Width:80.0
PSD name: **3dgrass**

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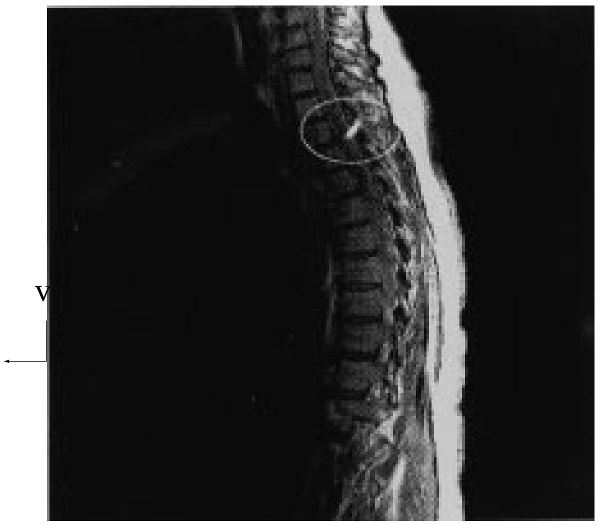
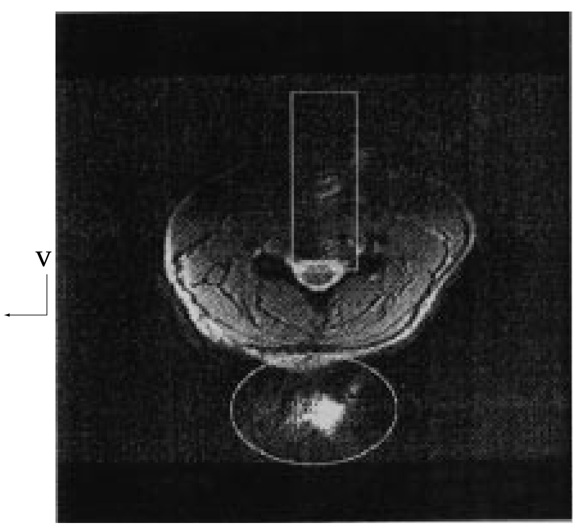
Startifact with FSE and 4 coil array;
Workaround: Don't swap phase and frequency.



|- - - - - - - - - - - - Image Parameters - - - - - - - - - - - -|
Image .... 15/28 Tr 3000.00 ms
Echo .... 1/2 Te 16.00 ms
Plane Sagittal Ti 0.00 ms
SPF Yes Contig..1 Flip 90 deg.
Acq 1/2 Fract Echo..2 Protocol: T SPINE
Slice Thickness. 3.00 mm NEX 2.00
Slice Spacing ... 1.00 mm Coil **CTLMID**
ScanTime . . 6:30.10 ETL.. 8 RawNo..35335
Bandwidth..32.0 Khz Ave Sar..0.99 Pk Sar..1.98
Scan Matrix.... 512 x 256 FOV **38.00 cm**
Scan Options: ST NP
SAT Pulses : SI
a:Loc:22.7:Width:75.0
PSD name: **fse**

Large Startifact and Annefact with CTLTOP
Workaround: Uses CS12 instead of CTLTOP to eliminate signals from coils 3-4 alaising into image.



|- - - - - - - - - - - - Image Parameters - - - - - - - - - - - -|
Image ..... 7 / 20 Tr 450. 00 ms
Echo .... 1 / 1 Te 15.00 ms
Plane Unknown Ti 0.00 ms
SPF....No Contig.. 1 Flip.... 20 deg.
Acq.... 1/1 Fract Echo..1 Protocol:CERVICAL SPINE
Slice Thickness. 3.00 mm NEX 3.00
Slice Spacing... 1.00 mm Coil **CTLTOP**
ScanTime.. 8:45.70 ETL.. 0 RawNo..3078
BandWidth..16.0 Khz Ave Sar..0.58 Pk Sar..1.16
Scan Matrix 256 x 192 FOV **20.00 cm**
Scan Options: FC ST
SAT Pulses : s:Loc:36.3:Width:80.0
i:Loc:6480.3:Width:80.0 a:Loc:33.7:Width:80.0
PSD name: **2dfast**

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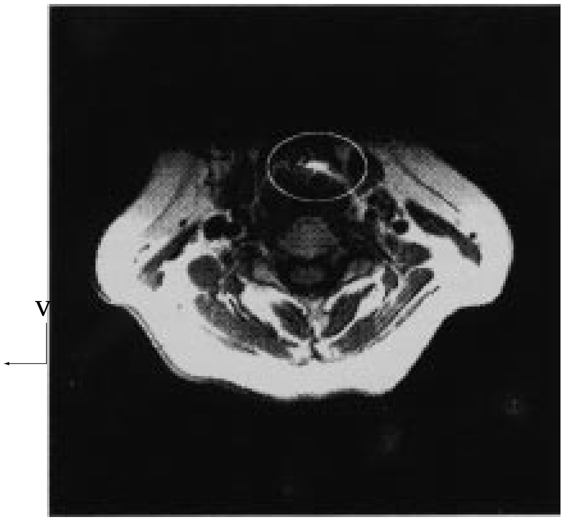
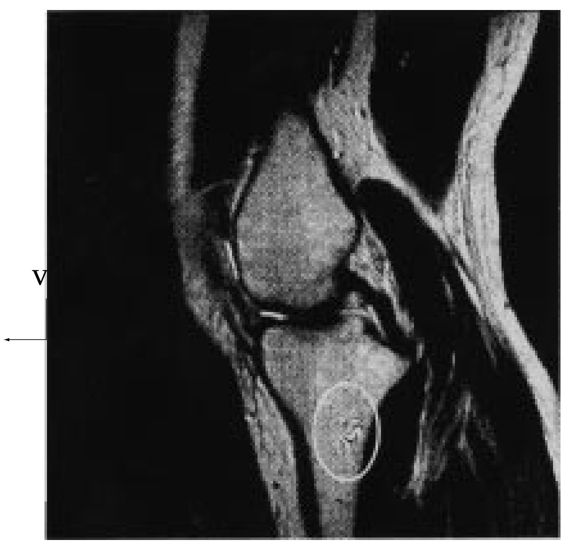
Startifact with MEMP (2d SE) scan using CTLTOP
Workaround: Use CS12 to prevent alaising from signals picked up by coils 3&4.



|- - - - - - - - - - - - Image Parameters - - - - - - - - - - - -|
Image ....4/ 2 Tr 800.00 ms
Echo ....1/ 1 Te 18.00 ms
Plane ......Unknown Ti 0.00 ms
SPF....No Contig.. 1 Flip.... 90 deg.
Acq.... 1/0 Fract Echo..0 Protocol:
Slice Thickness. 5.00 mm NEX 3.00
Slice Spacing... 1.00 mm Coil **CTLTOP**
ScanTime.. 7:50.45 ETL.. 0 RawNo..63491
BandWidth.. 16.0 Khz Ave Sar. .0.68 Pk Sar. 1.35
Scan Matrix 256 x 192 FOV **20.00 cm**
Scan Options: ST
SAT Pulses : i: Loc:6509.6:Width:0.0
PSD name: **memp**

Startifact with Fast Spin Echo in the Extremity Coil
Workaround: Don't swap phase and frequency.



|- - - - - - - - - - - - Image Parameters - - - - - - - - - - - -|
Image 14/32 Tr 3400.00 ms
Echo 2/2 Te 115.00 ms
Plane Unknown Ti 0.00 ms
SPF....**Yes** Contig.. 1 Flip.... 90 deg.
Slice Thickness. 4.00 mm NEX 2.00
Slice Spacing... 1.00 mm Coil **EXTREM**
ScanTime.. 7:22.10 ETL.. 8 RawNo..2566
BandWidth.. 16.0 Khz
Scan Matrix 256 x 256 FOV **16.00 cm**
Scan Options: NP
SAT Pulses : None
PSD name: **fse**

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John A. Johnson
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